NLP Risk Engine + Index Rebalancer + Stress Tester

Turning news & social chatter into structured risk signals

Candidate: Mehwish | College: VIT | Solo submission

Problem

• Market-moving information arrives as unstructured text (news, X posts) faster than analysts can read it.

• Risk and portfolio teams need it as numbers they can act on: how positive/negative, what kind of event, how severe.

Approach

• One NLP Risk Engine: news + social text in, one signal per text out: sentiment (-1..1), event type, impact (1..10).

• Signals published via file, REST API and an in-process pub/sub bus.

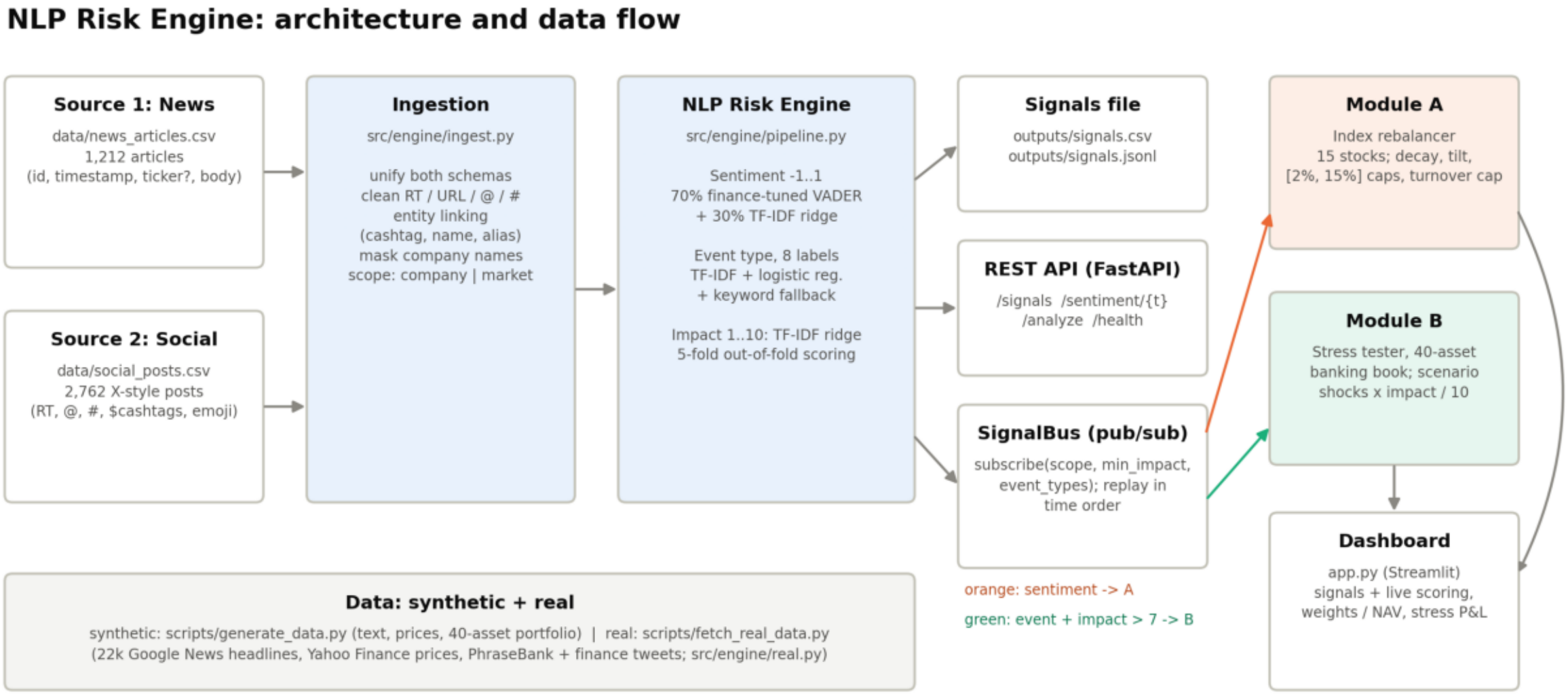
• Both downstream modules built on the bus:

- A: sentiment-tilted index of 15 S&P 100 stocks

- B: event-triggered stress test of a 40-asset wholesale banking book

• Built on synthetic data, then tested on REAL public data: 22,442 Google News headlines, Yahoo Finance prices, 4,652 labelled real texts.

System design and data flow



Implementation: the NLP Risk Engine

• Ingestion: 2 sources -> one schema; social cleaning (RT, URLs, @, #); entity linking by cashtag/name/alias; company names masked so models read language, not tickers.

• Sentiment: finance-tuned VADER lexicon + TF-IDF model. Event: TF-IDF + logistic regression (8 labels), keyword fallback when unsure. Impact: TF-IDF ridge.

• Synthetic corpus scored out-of-fold (5-fold): sentiment r 0.908, event accuracy 0.969, but test text shares the generator's templates, so this is the optimistic number.

• Real text exposed the gap: the synthetic-trained sentiment model is no better than always guessing 'neutral'. Retraining on real labelled text fixed it.

• Tech: Python 3.13, pandas, scikit-learn, vaderSentiment, FastAPI, Streamlit, Plotly; 33 pytest tests.

Sentiment test set	n	Synthetic-trained	Real-trained
Real tweets (held-out)	2,388	acc 0.59 / F1 0.52	acc 0.83 / F1 0.77
Real news (PhraseBank, 5-fold)	2,264	acc 0.61 / F1 0.49	acc 0.89 / F1 0.86
'Neutral' baseline: tweets / news		acc 0.66 / 0.61	
Handwritten holdout: event acc.	20	0.90	

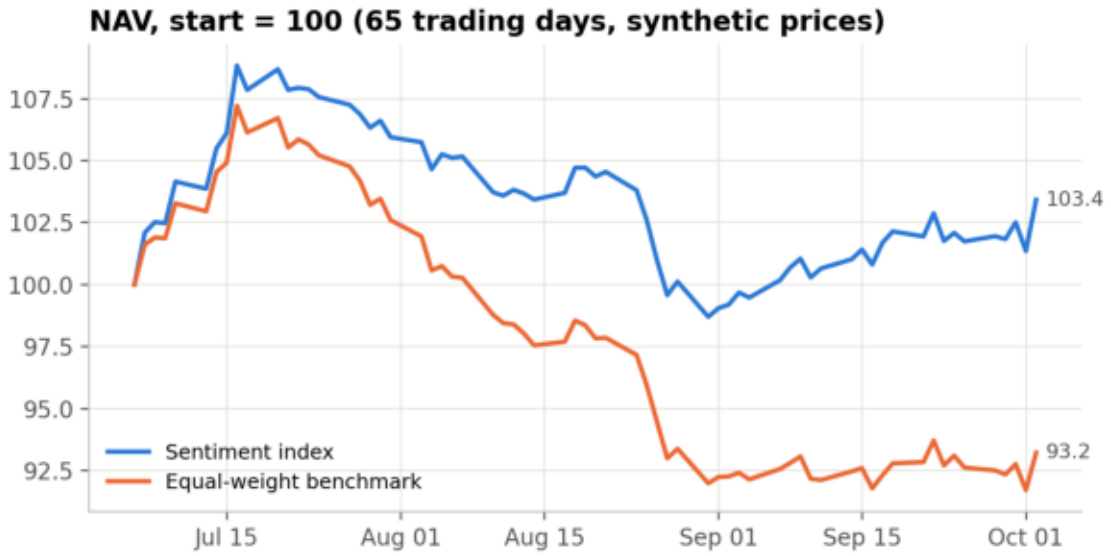
Lesson: a model that looks excellent on synthetic data can fail on real wording. Measuring on held-out real text (official splits; blend weight chosen by CV on training data only) is what made the engine credible.

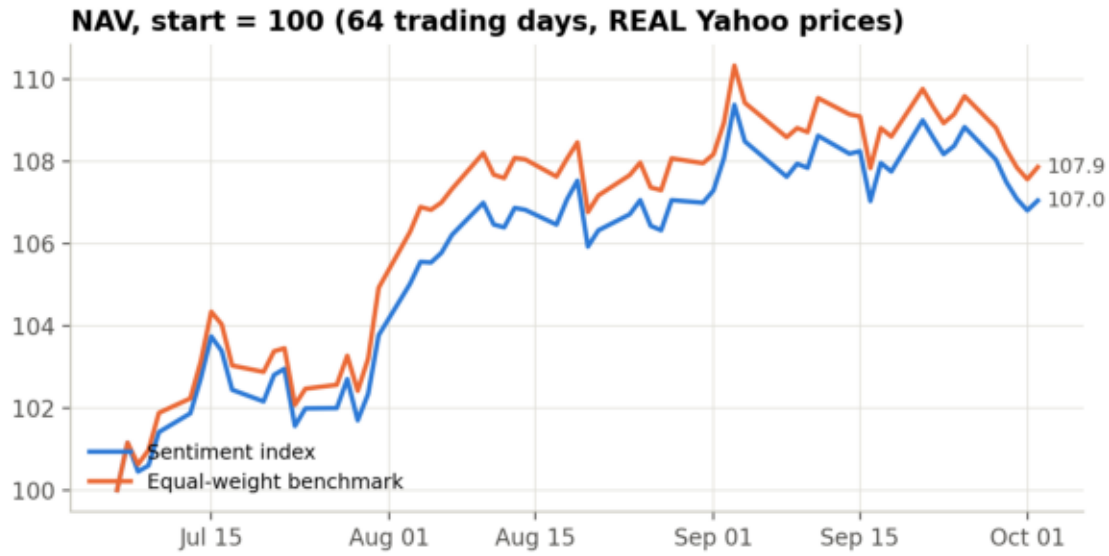
Benchmark vs FinBERT (pre-trained finance transformer)

	FinBERT (off the shelf)	Our real-trained model
Real tweets, held-out (2,388)	acc 0.71 / F1 0.66	acc 0.83 / F1 0.77
Speed, same CPU	32 texts/s	10,921 texts/s
Model size	438 MB	3.4 MB

Ours is more accurate on social text and ~342x faster. FinBERT's 0.97 on PhraseBank is in-sample (it was trained on it), so it is not compared.

Results, Module A: sentiment-driven index rebalancing



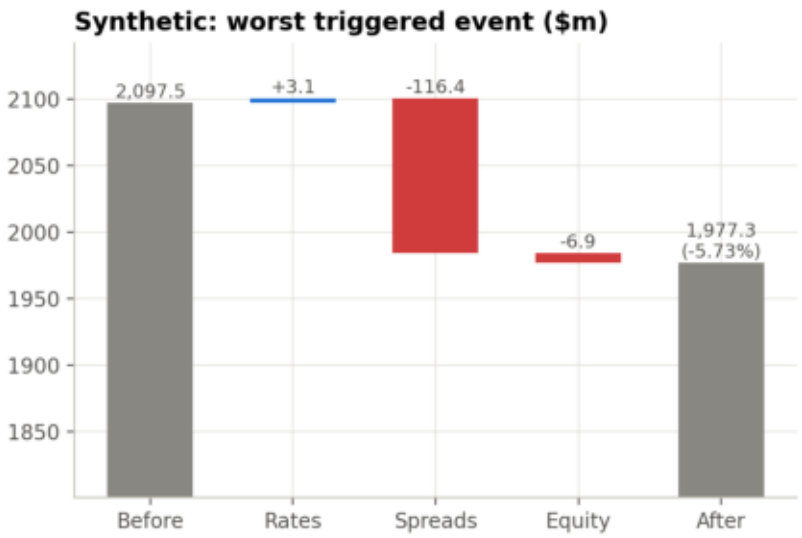


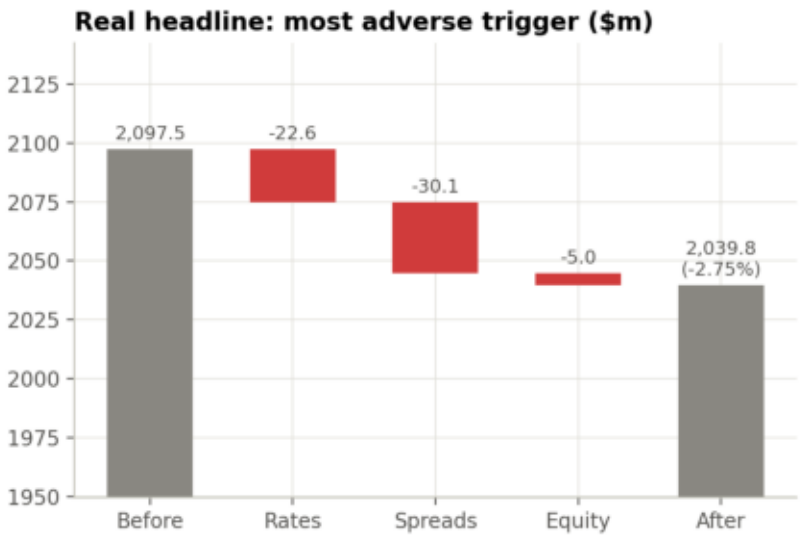
Data	Sentiment index	Equal weight	Shuffled-sentiment placebo	Sharpe (index / EW)
Synthetic text + synthetic prices	+3.4%	-6.8%	-6.9%	1.06 / -2.08
REAL headlines + REAL prices	+7.0%	+7.9%	+7.8% (+7.0%..+8.4%)	2.66 / 2.89

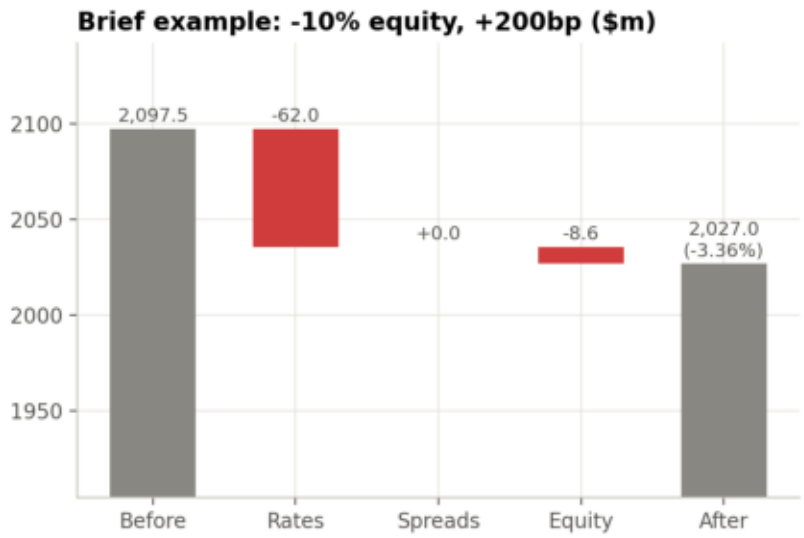
• Rule: decayed (36h), impact x source-weighted sentiment -> weight = 1/15 x exp(1.5 x score), bounds [2%, 15%], max 20% turnover/day, weights set at close t earn t+1 (no look-ahead; real headlines enter the next day).

• Synthetic edge is circular (prices built from the same events). On real data the tilt does NOT beat equal weight: public headlines are priced in fast. Reported as-is, no re-tuning on the test period.

Results, Module B: event-triggered stress testing







• Trigger = impact > 7, confident event label (>= 0.60), adverse sentiment, 24h cooldown per event+ticker. Synthetic: 369 high-impact signals -> 238 tests. Real: 561 -> 51 (the confidence rule removed 458 false alarms, e.g. a lifestyle story about a 'default setting').

• Worst synthetic: Credit Event on GS (impact 10) -5.73% = -$120.2m on a $2,098m book. Real: 'US economy grows sluggish 1.5% in second quarter as inflation tops Fed target' -> -2.75%.

• Brief's example (-10% equity, +200bp): -3.36%; swaps and puts offset $+77.8m. Valued on market value instead of risk notional the hedge shows only $+1.5m: the naive approach mis-states risk.

Domain impact and challenges

Domain impact

• Portfolio managers: a transparent, rules-based tilt that reacts within one close to news flow, with turnover and concentration limits a risk committee can sign off.

• Risk / treasury desks: every high-impact headline becomes an immediate, explainable 'what does this do to our book' number, by rates / credit / equity, asset type and sector.

• On real news it flagged the US-Iran war, Russia sanctions, the US-Canada trade war and Fed rate hikes as stress events.

• One engine, many consumers: new modules subscribe to the bus without touching the NLP code.

Challenges

• Synthetic-to-real gap: great synthetic scores, coin-flip on real text -> retrained on real labelled data.

• Alert storms and false alarms: strict threshold, confidence and adverse-only rules, cooldown.

• Look-ahead: signals after 16:00 roll to the next close; real headlines enter the next day; tested explicitly.

• Data access: GDELT rate-limited us (HTTP 429), so we used Google News RSS for the same publishers.

• Derivatives: P&L on risk notional, not ~0 market value; sign tests for swaps, CDS, puts.

Limitations and next steps

Limitations (stated plainly)

• Event and impact models are trained on synthetic labels only; on real text impact keys on dramatic words.

• Real backtest covers 64 trading days and 15 stocks: far too short to claim (or rule out) alpha.

• The banking portfolio is synthetic; Google News timestamps are often date-only (handled conservatively).

• Stress model is first-order (duration / beta): no convexity, vega, correlations, rating migration or default loss.

• Scenario sizes are expert judgement, not calibrated to history.

Next steps

• Label real headlines for event type and learn impact from actual price reactions (event study).

• Fine-tune FinBERT on real tweets (off the shelf it loses: 0.71 vs our 0.83, ~340x slower); use it on high-impact signals only.

• Longer real backtest with intraday prices and transaction costs; test faster signal decay.

• Calibrate shocks to historical episodes; add convexity, correlated factor shocks and credit migration.

• Replace the in-process bus with Kafka and stream live news into the dashboard.